Timeouts retries and cancellation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Timeouts retries and cancellation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Approval gates and idempotency

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

State machines for bounded agents

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Timeouts bound waiting; retries consume a finite budget. **Cancellation** should stop future work while accounting for operations already committed.

Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds.

Practise with fading help

> Complete the supported Timeouts retries and cancellation trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Timeout**s retries and **cancellation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Timeouts retries and cancellation explanation against the supplied facts before discussing it.

Use the worked case for **Timeout**s retries and **cancellation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Timeouts retries and cancellation facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Timeouts retries and cancellation facts and keep your reasoning.

Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Timeouts retries and cancellation facts and keep your reasoning.

A report omits this qualification: A **timeout** means the caller lacks a timely result, not necessarily that the remote operation did not occur.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Timeouts retries and cancellation facts and keep your reasoning.

Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Timeouts retries and cancellation facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Timeouts retries and cancellation to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **timeout**s retries and **cancellation evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **timeout**s retries and **cancellation**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Timeouts retries and cancellation

> Retry within declared limits. A read-only lookup allows two attempts of at most one second each. Cancellation forbids further attempts. Explain the labelled step before checking the changed condition.

Retry within declared limits. A read-only lookup allows two attempts of at most one second each. **Cancellation** forbids further attempts.

First attempt times out at 1 second. Second succeeds in 0.4 seconds. Ignore overhead. No **cancellation** occurs.

AI ACADEMY / AI-6/16

Timeouts retries and cancellation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Bound first wait

Stop attempt one after its one-second allowance.



2. Check retry

One attempt remains, and cancellation has not occurred.



3. Complete next attempt

Accept the checked record; total waiting is 1.4 seconds.

CHANGE ONE THING

Cancellation arrives after the first timeout.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate waiting time for a one-second **timeout** followed by a 0.4-second success. What changes if **cancellation** arrives after the first attempt but before the retry?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Retry within declared limits. A read-only lookup allows two attempts of at most one second each. **Cancellation** forbids further attempts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept the checked record; total waiting is 1.4 seconds.

> 1. Complete next attempt

2. Bound first wait

3. Check retry

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Complete next attempt Accept the checked record; total waiting is 1.4 seconds. A **timeout** bounds waiting; a retry tries again; **cancellation** withdraws the request.

They are separate conditions on the next action.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: **Cancellation** arrives after the first **timeout**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Do not start the second attempt.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Do not start the second attempt. **Cancellation** overrides the unused retry allowance.

Repair a missing building block

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

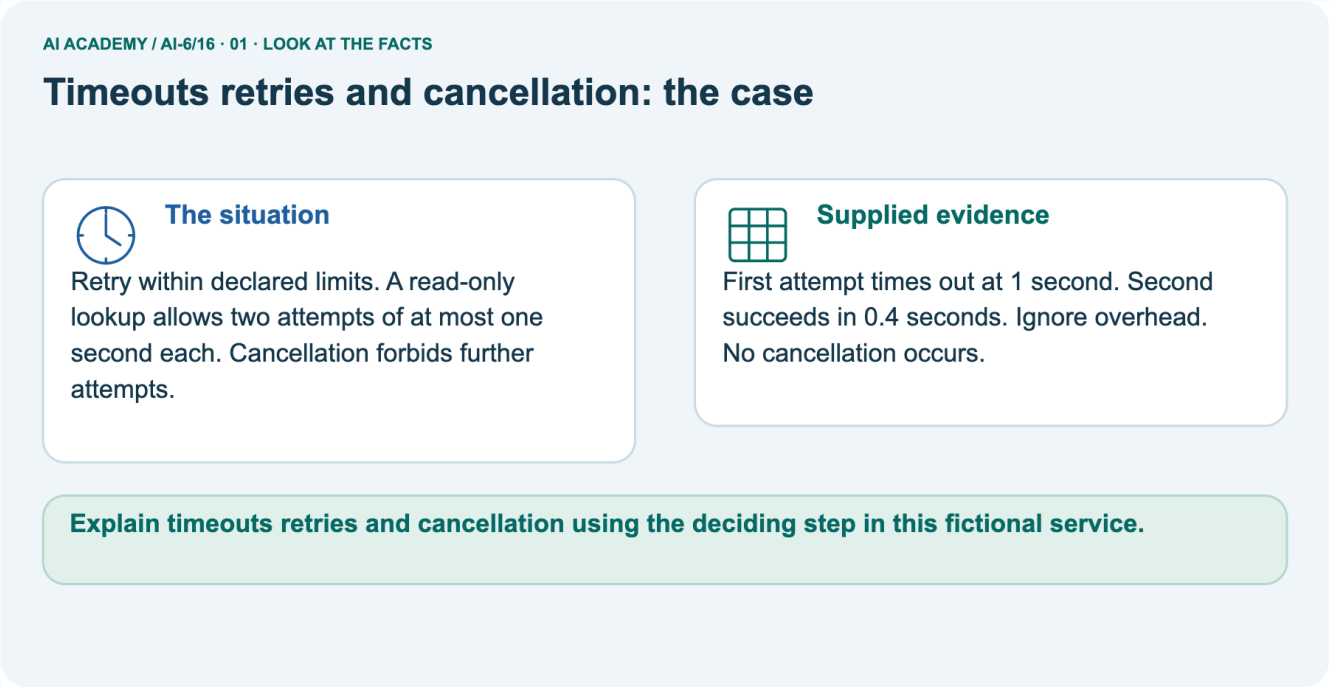
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Timeouts retries and cancellation picture from its labels, then record a relevant limit.

01. Timeouts retries and cancellation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

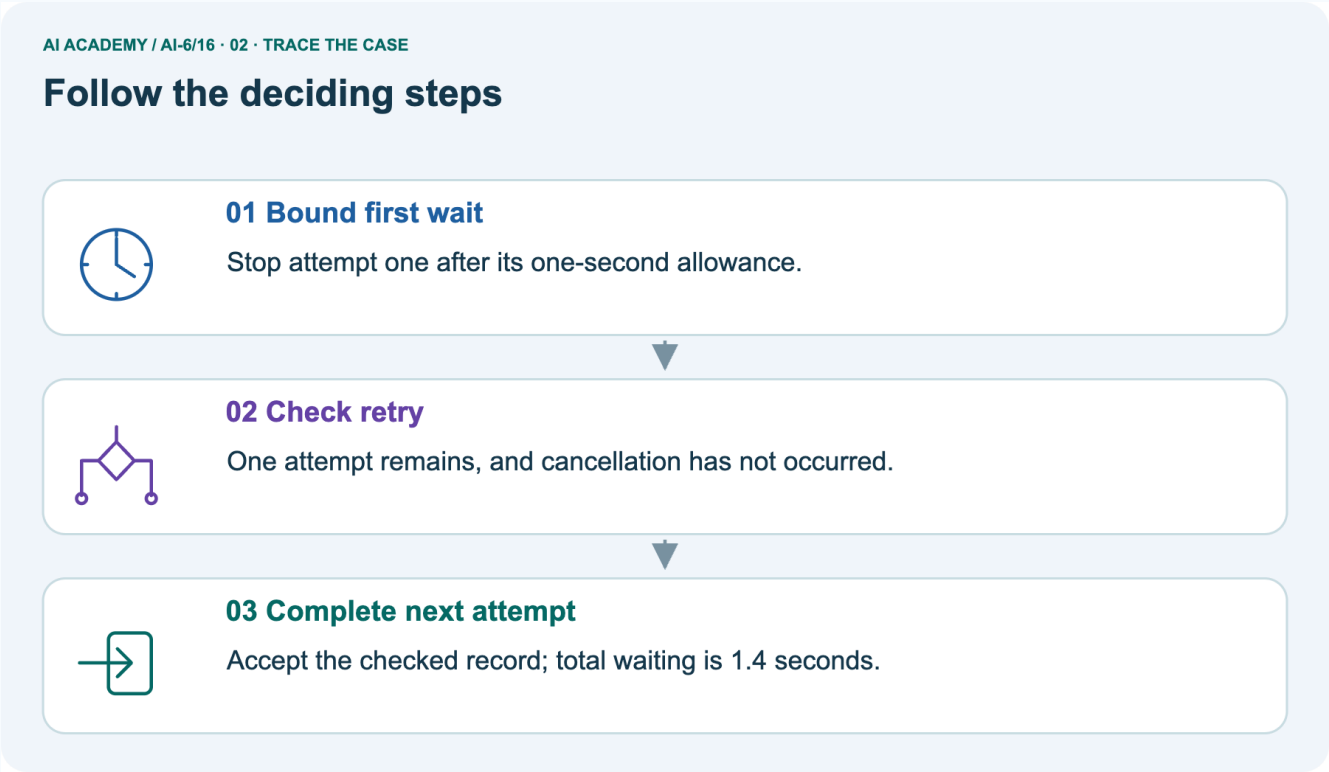
Retry within declared limits. A read-only lookup allows two attempts of at most one second each. **Cancellation** forbids further attempts. Supplied facts: First attempt times out at 1 second. Second succeeds in 0.4 seconds.

Ignore overhead. No **cancellation** occurs.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



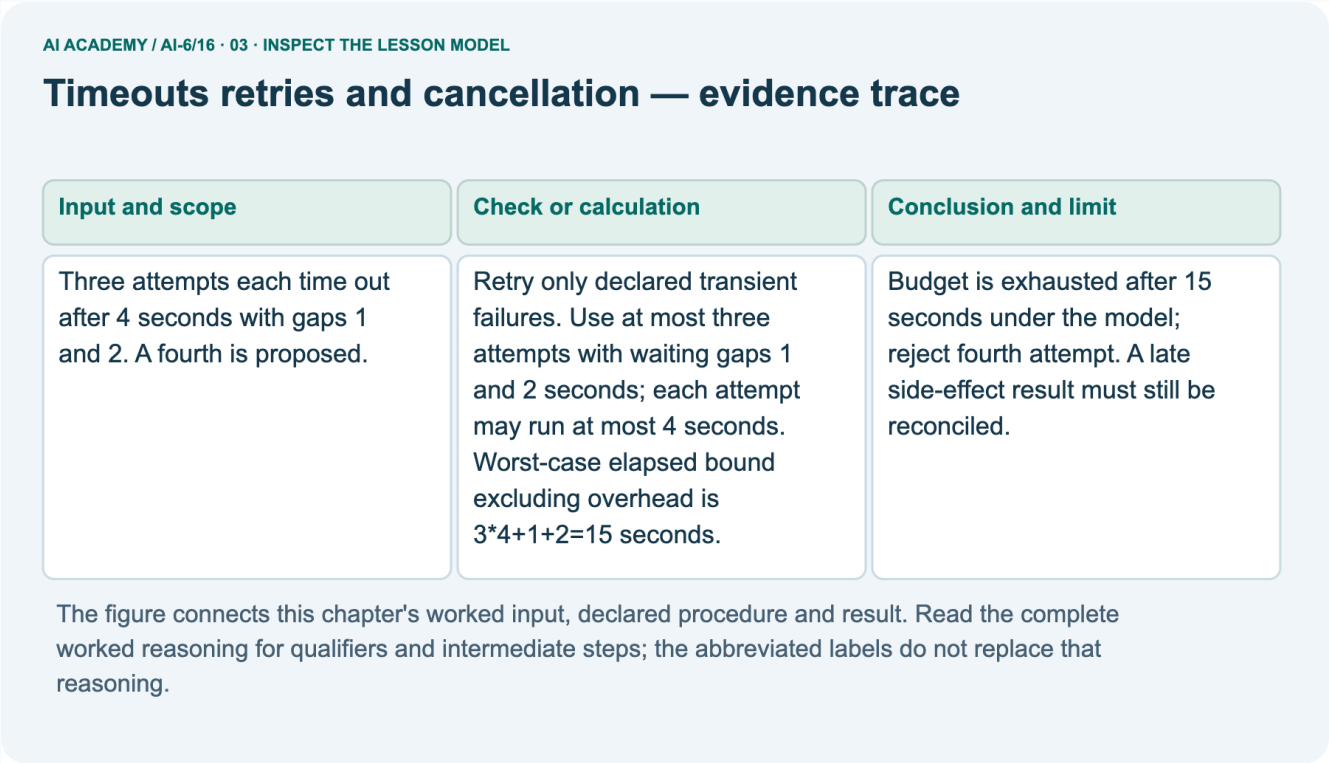
Follow the labelled picture with the matching explanation. Open the vector diagram

Bound first wait: Stop attempt one after its one-second allowance. → Check retry: One attempt remains, and **cancellation** has not occurred. → Complete next attempt: Accept the checked record; total waiting is 1.4 seconds.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Timeouts retries and cancellation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Timeouts retries and **cancellation** — **evidence** trace. Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. | Retry only declared transient failures.

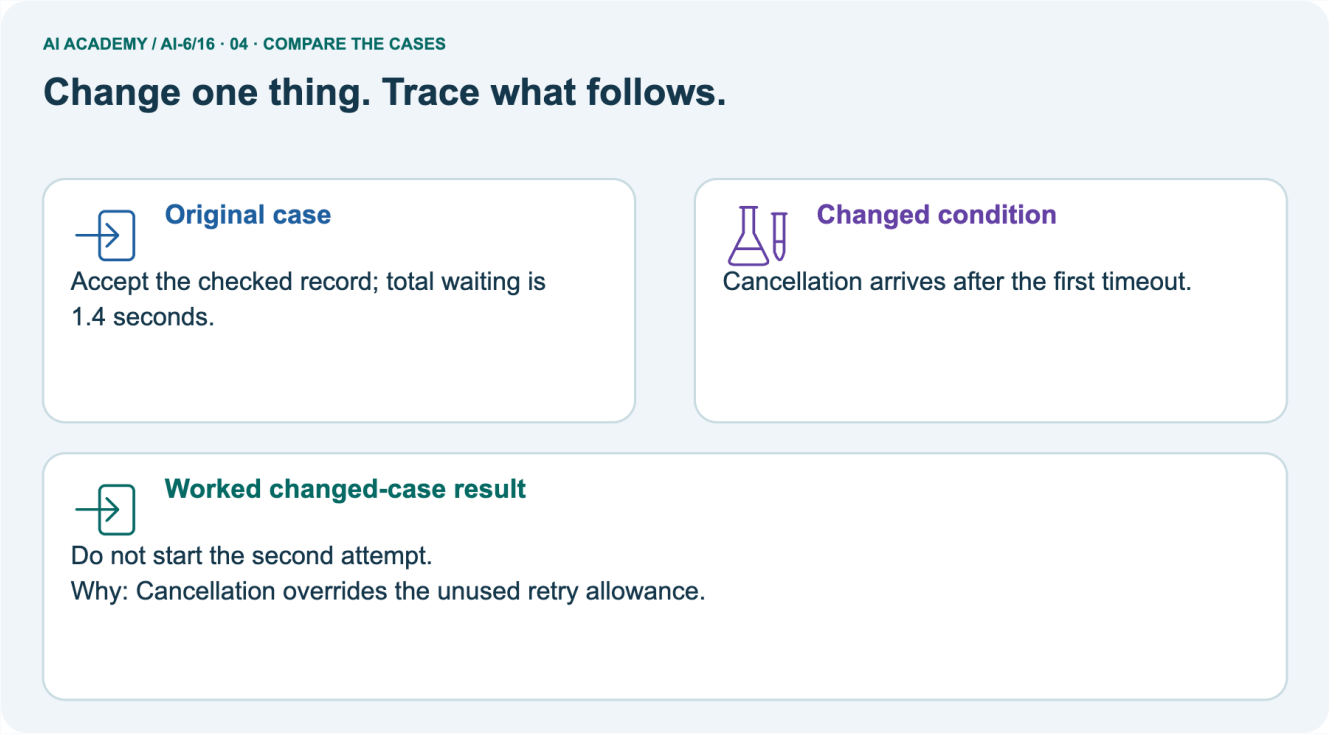
Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds.

Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds. | Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

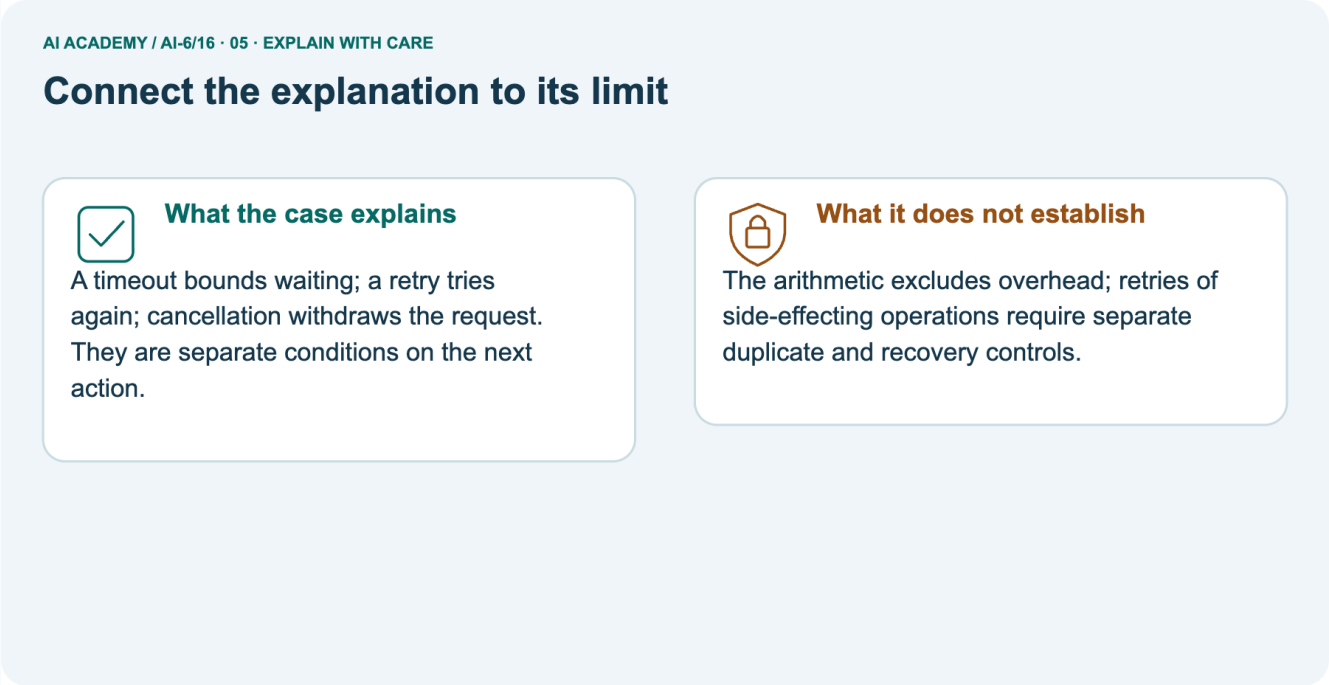
Original case: Accept the checked record; total waiting is 1.4 seconds. Changed condition: **Cancellation** arrives after the first **timeout**. Expected result: Do not start the second attempt.

Reason: **Cancellation** overrides the unused retry allowance.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

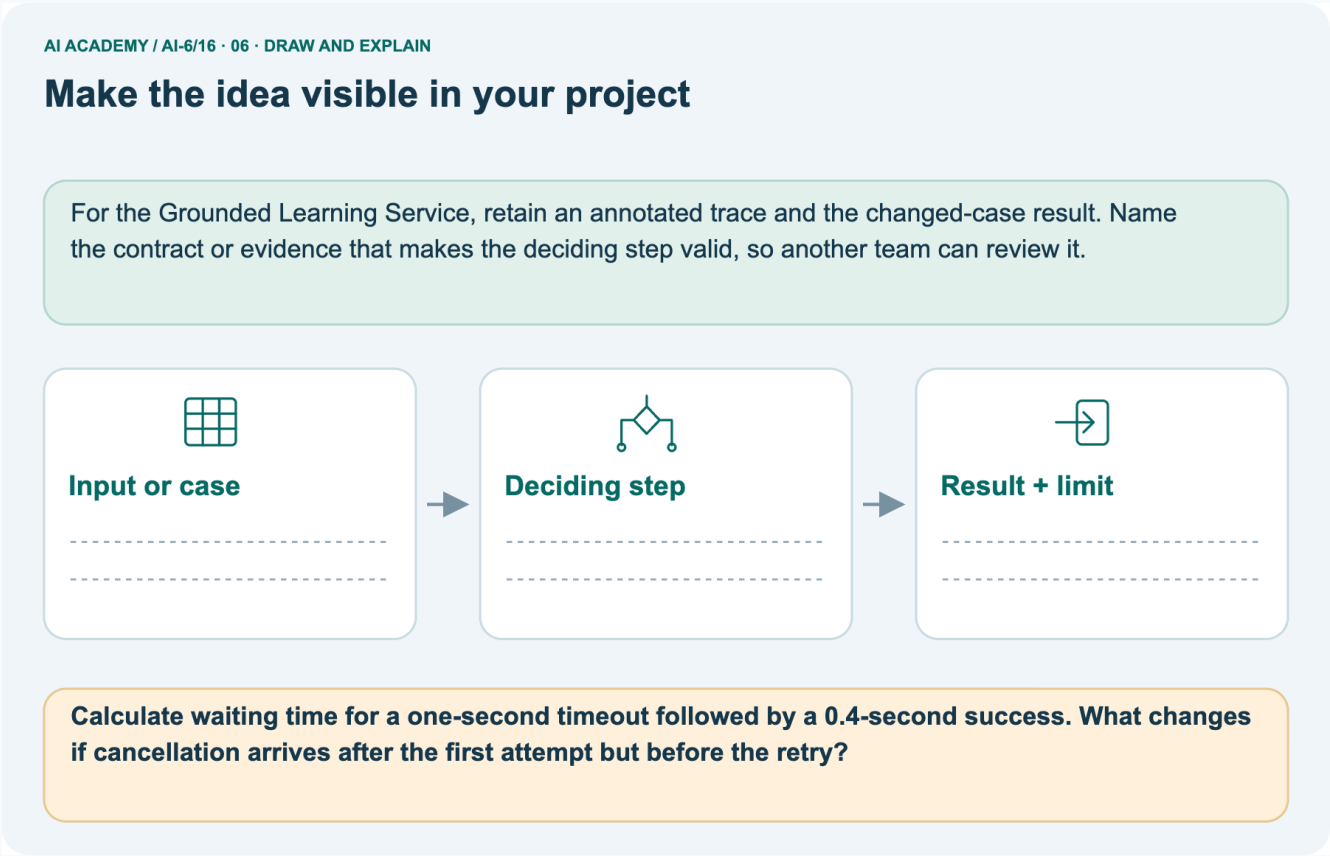
A **timeout** bounds waiting; a retry tries again; **cancellation** withdraws the request. They are separate conditions on the next action.

Limit: The arithmetic excludes overhead; retries of side-effecting operations require separate duplicate and recovery controls.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate waiting time for a one-second **timeout** followed by a 0.4-second success.

What changes if **cancellation** arrives after the first attempt but before the retry?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.